Algebra 2

Practice Test — Study Copy

Unit 3: Polynomial Functions

Name: ____________________________ Date: ________________ Period: _________

Keep in Mind

This is a practice test. It mirrors the real Unit 3 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

1. Degree _______
2. Leading coefficient _______
3. End behavior _______
4. Multiplicity of a zero _______
5. Turning point _______
6. Factor Theorem _______
7. Rational Root Theorem _______
8. Fundamental Theorem of Algebra _______

(A) The number multiplying the highest-power term; its sign decides the right-hand arm.

(B) The greatest exponent on the variable once the polynomial is in standard form.

(C) How many times the factor $(x-r)$ appears — odd means the graph crosses at r, even means it touches.

(D) A point where the graph changes from increasing to decreasing, or the reverse.

(E) What the graph does at its far left and far right, as $x \to \pm\infty$.

(F) $(x - r)$ is a factor of $f(x)$ exactly when $f(r) = 0$.

(G) Every rational zero of a polynomial with integer coefficients looks like $\pm\frac{p}{q}$, where p divides the constant term and q divides the leading coefficient.

(H) A polynomial of degree $n \geq 1$ has exactly n complex zeros, counted with multiplicity.

Part B — Multiple Choice (12 pts)

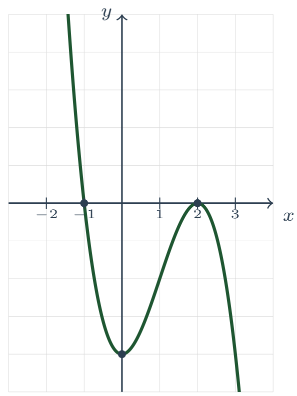
Circle the best answer.

1. The end behavior of $f(x) = -2x^4 + 3x^2 - 1$ is (A) both arms up (B) both arms down (C) left down, right up (D) left up, right down
2. $(2x - 5)^2 =$ (A) $4x^2 - 25$ (B) $4x^2 + 25$ (C) $4x^2 - 20x + 25$ (D) $4x^2 - 10x + 25$
3. Factored completely, $x^3 + 27$ is (A) $(x + 3)(x^2 - 3x + 9)$ (B) $(x + 3)(x^2 + 3x + 9)$ (C) $(x + 3)^3$ (D) prime
4. The remainder when $f(x) = x^3 - 4x^2 + 1$ is divided by $x - 2$ is (A) -7 (B) -23 (C) 7 (D) 0
5. Which value is *not* a possible rational zero of $f(x) = 3x^3 + 2x^2 - x + 4$? (A) -2 (B) $\frac{4}{3}$ (C) $\frac{3}{4}$ (D) 1
6. A polynomial with *real* coefficients has degree 4 and zeros 2, -1, and $3i$. Its fourth zero must be (A) $-3i$ (B) $3i$ (C) -3 (D) $\frac{1}{3}$

Part C — Short Answer & Computation (40 pts)

Show your work.

1. **Read the graph.** The polynomial function f is graphed below. Each gridline is one unit.



(a) Degree: even or odd? ________ Leading coefficient: positive or negative? __________

(b) As $x \to -\infty$, $f(x) \to$ ________; as $x \to +\infty$, $f(x) \to$ ________

(c) Zeros, each with its multiplicity, and whether the graph crosses or touches there: ____________________________________

(d) y-intercept: __________; number of turning points: _______

(e) Relative maximum: ___________; relative minimum: ___________;

absolute maximum or minimum: ______________________

(f) Interval(s) where f is increasing: ______________; domain: ___________; range: ___________

2. **Operations.** Simplify. Write each answer in standard form.

(a) $(3x^3 - 5x + 2) - (x^3 + 4x^2 - 5)$

(b) $(2x - 3)(x^2 + 4x - 1)$

(c) For your answer to (b), give the degree ______, the leading coefficient ______, and the end behavior ________________________

3. **Factor completely.** Every factor must be prime over the integers.

(a) $2x^3 - 18x$ (b) $27x^3 - 8$ (c) $x^3 + 3x^2 - 4x - 12$

4. **Division and the Remainder Theorem.**

(a) Divide using *long division* and write the answer as $Q(x) + \dfrac{R}{D(x)}$:

$$(x^3 + 2x - 9) \div (x - 2)$$

(b) Use the *Remainder Theorem* to find the remainder when $g(x) = x^3 + 2x^2 - 5x + 6$ is divided by $x + 3$. Is $x + 3$ a factor of g? Explain in one sentence.

5. **Synthetic division.** You are told that $x = 3$ is a zero of

$$f(x) = x^3 - 2x^2 - 5x + 6.$$

Use synthetic division to divide out that zero, factor f completely, and list *all* of its zeros.

6. **Rational Root Theorem.** For $f(x) = 2x^3 - x^2 - 8x + 4$:

(a) List every possible rational zero.

(b) Find all real zeros of f, showing the test and the division you used.

7. **Solve over the complex numbers.** $x^3 - 3x^2 + 4x - 12 = 0$

(a) Before solving: how many solutions must this equation have, counted with multiplicity, and which theorem says so? ________________________________

(b) Solve. Give every solution.

(c) How many of your solutions are real, and how many x-intercepts does the graph of $y = x^3 - 3x^2 + 4x - 12$ therefore have? ___________________________

8. **Build a polynomial from its zeros.** Write a polynomial function of *least degree* with *real* coefficients whose zeros are -1 and $2i$. Give your answer in factored form *and* in standard form.

Part D — Extended Response (12 pts)

Explain your reasoning in full sentences.

1. **The graph plan.** Consider $f(x) = -2(x+2)(x-1)^2(x-3)$. Do *not* expand it. **(a)** State the degree and the leading coefficient, and explain how you got each one from the factored form. What end behavior do they force? **(b)** List the zeros with their multiplicities, and say whether the graph crosses or touches the x-axis at each. **(c)** Find the y-intercept. **(d)** Build a sign chart and state every interval on which $f(x) > 0$. **(e)** At most how many turning points can this graph have, and why?

2. **Modeling.** An open-top box is made from a 12 in by 6 in sheet of cardboard by cutting a square of side x from each corner and folding up the sides. **(a)** Write the volume $V(x)$ in factored form, then expand it into standard form. **(b)** State the feasible domain of the model and justify it from the sheet's dimensions. **(c)** Find $V(1)$ and state its units. **(d)** Check that $x = 6$ makes $V(x) = 0$. Explain why 6 is a genuine zero of the polynomial but not a possible cut length.